

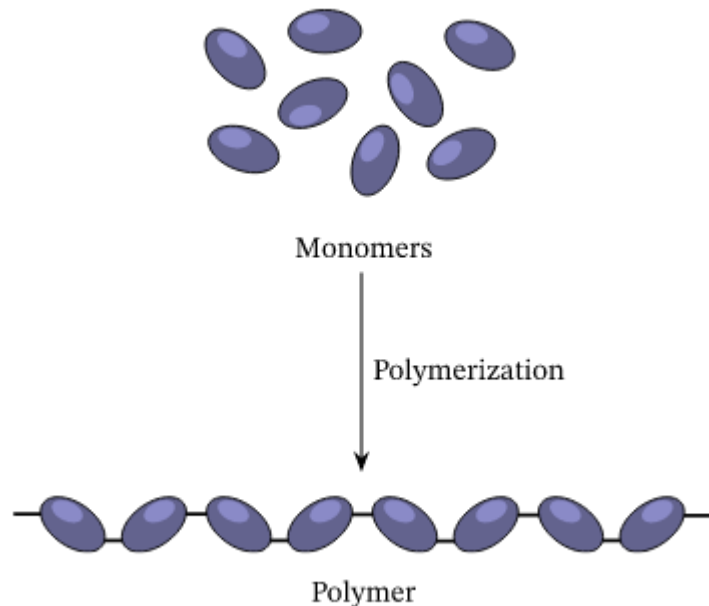
Polymeric Materials I

- **Introduction.**
- **Classification**
- **Polymerization Mechanisms**
- **Molecular weight**
- **Structural aspects**
- **Crystallinity**
- **Thermal transitions: T_m and T_g .**
- **Mechanical properties**

INTRODUCTION: What is a Polymer?

Monomer and Polymer

A **monomer** is a **single atom**, small molecule, or molecular fragment.



Polymer: Organic compound, natural or synthetic, with high molecular weight made of repetitive structural units.

Large size chains formed by the **covalent union** of various **monomer** units (macromolecule)

Etimology: Polymer $\Rightarrow$ (“πολυ” = “many”) + (“μερος” = “part”) poly-mers (many equal parts)

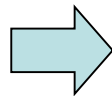
INTRODUCTION: DEFINITION

IUPAC: Polymer $\Rightarrow$ *A substance composed of macromolecules or molecules of high relative molecular mass, the structure of which essentially comprises the multiple repetition of units derived, actually or conceptually, from molecules of low relative molecular mass”.*

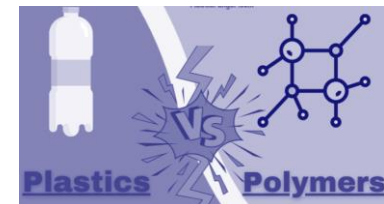
Is a plastic the same as a polymer?

Polymers can occur naturally or artificially produced, consisting of chains of monomers or individual molecules linked together. **Plastics**, on the other hand, are a **specific type of polymer** made up of chains of polymers that can be partly organic or entirely synthetic. Simply put, all **plastics are considered polymers**, but not all polymers can be classified as plastics.

Plastics



Polymers + additives (fillers, lubricants, ...)



INTRODUCTION

Origins of Humanity $\Rightarrow$ **natural polymers**: leather, wool, cotton, natural rubber cellulose, starch...

"Carbs" like spaghetti are natural polymers



Wood is a natural polymer used for paper



Crustacean shells are made of chitin, a natural polymer



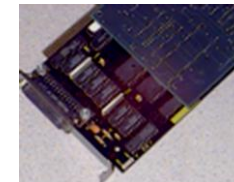
natural polymers



Natural latex from rubber tree is a polymer



Synthetic polymers



Applications: products with high consumption, pneumatics, packaging
Automobiles, aeronautics, construction, coatings for wires, Tubes, boxes ect.

INTRODUCTION: HISTORICAL DEVELOPMENT

1838 Vulcanizing of natural rubber



Charles Goodyear



1846 Cellulose nitrate

1870 Celluloid- made by nitration of cellulose



1907 Bakelite

1920 Macromolecular Hypothesis (Staudinger)



1953 Nobel Prize

1926 Polyvinyl chloride (PVC)

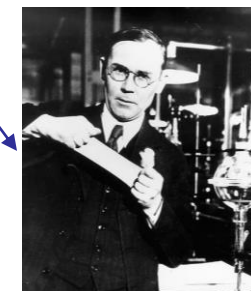
1933 Poly ethylene (PE)

1938-39 Nylon (fibers) and Polystyrene (PS)

1954 Polypropylene (PP)

1960 Applications of Epoxy Resins

198- Polymers of high specifications

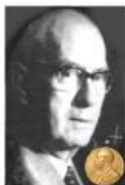


Wallace Hume Carothers,
Julian Hill, and others



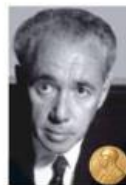
P. J. Flory

1963 Nobel Prize



K. Ziegler

1974 Nobel Prize



G. Natta



Alan J. Heeger



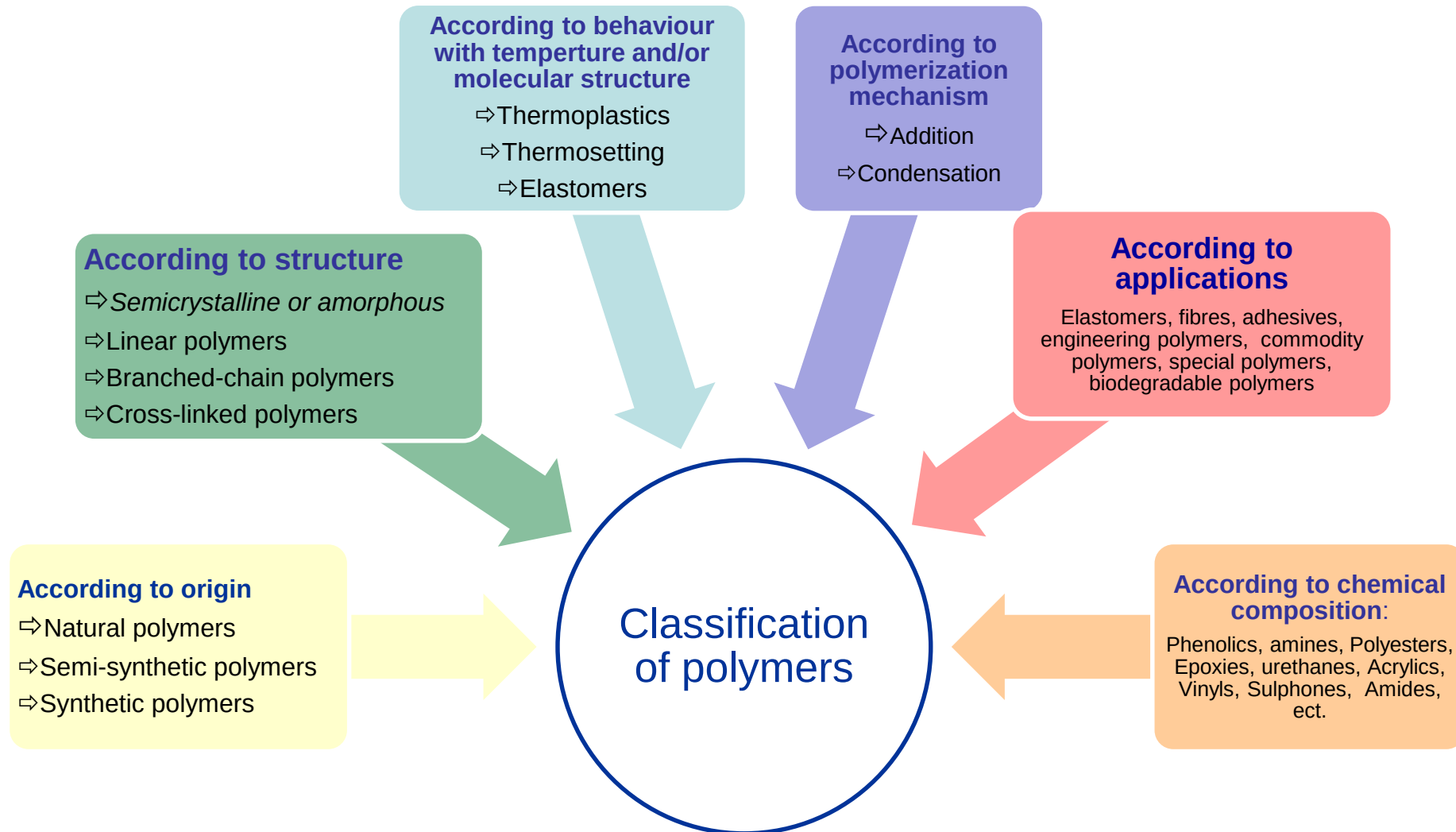
Alan G. MacDiarmid



Hideki Shirakawa

2000 Nobel Prize in chemistry

CLASIFICATION

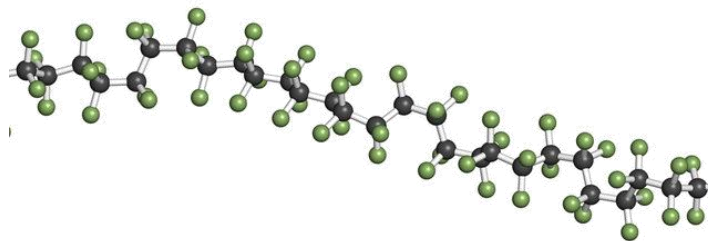


INTRODUCTION

Which factors determine the final properties of a polymer?

Structural aspects must be considered (**Molecular features**):

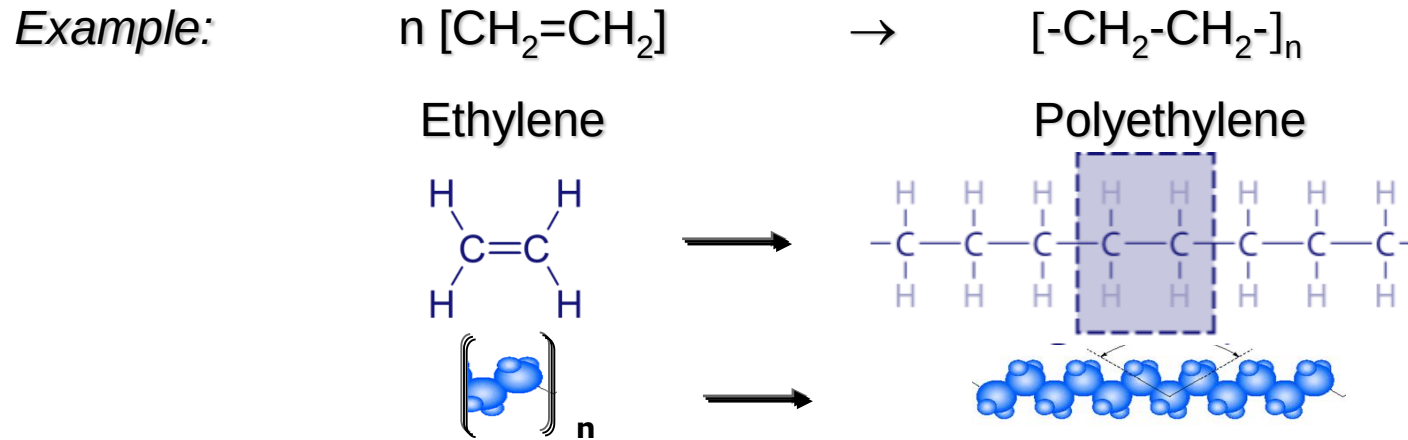
- **Monomeric nature** (families of polymers)
- **Number** of monomeric units (molecular weight)
- **Proportion** of chains with different molecular weight (distribution)
- **Monomeric functionality** (branching and cross-linking)
- **Relative position** of groups (tacticity and conformational changes)
- **Sorting** of units (sequences)
- **Orderly positioning** of chain sections (crystallinity)



POLYMERIZATION MECHANISM

How do you get a polymer?

Polymerization, any process in which relatively small molecules, called monomers, combine chemically to produce a very large chainlike or network molecule, called a polymer.



Types of polymerization?

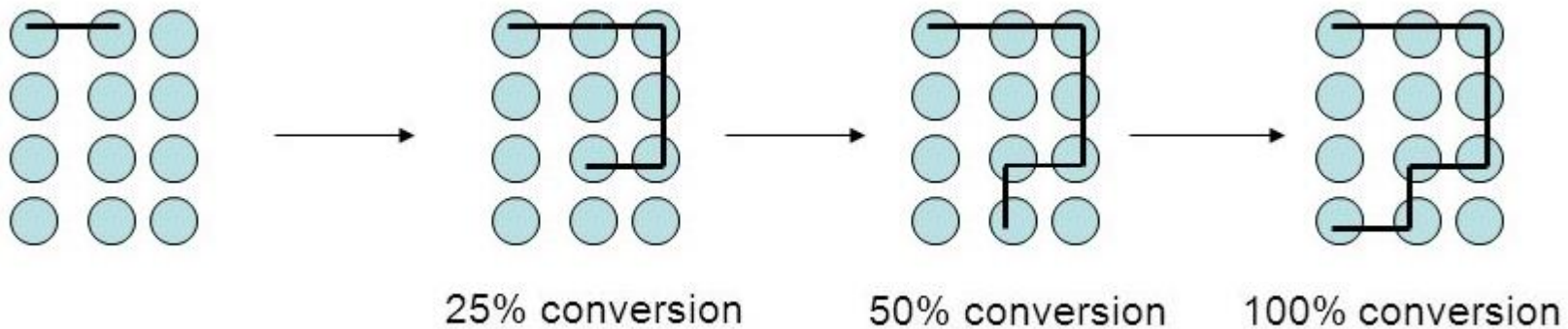
Chain-reaction (or addition) polymerization - Process by which polymer chains are built up by chemically combining monomers into long chain molecular polymers.

Step-reaction (or condensation) - A polymerization mechanism in which monomers chemically react with each other to produce polymers (often a small molecule e.g., water, methanol, etc. is condensed out as a byproduct)

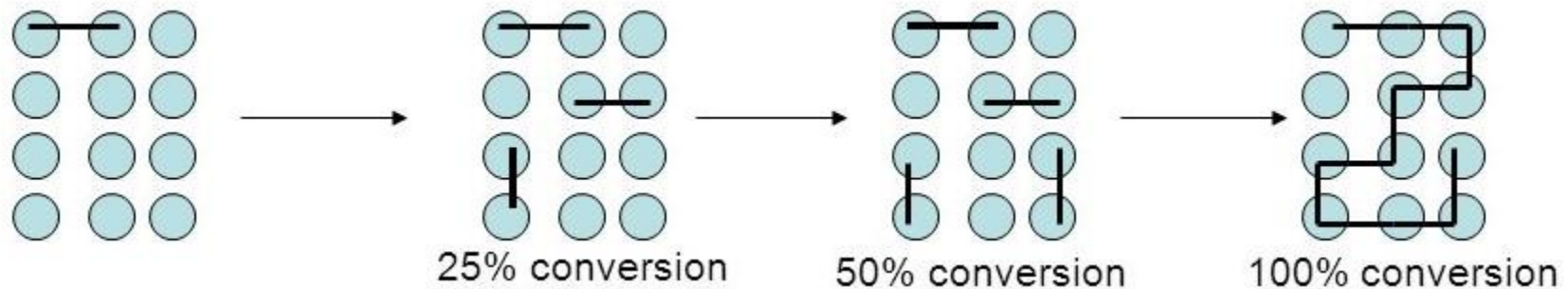
POLYMERIZATION MECHANISM

Synthesis of polymers

Chain-reaction (or addition)



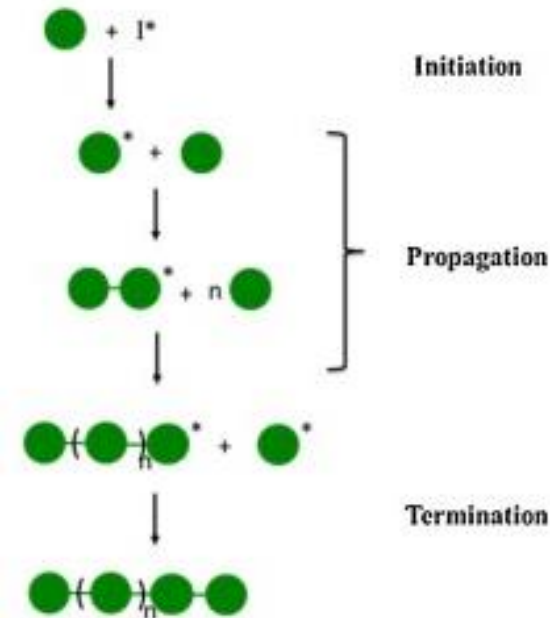
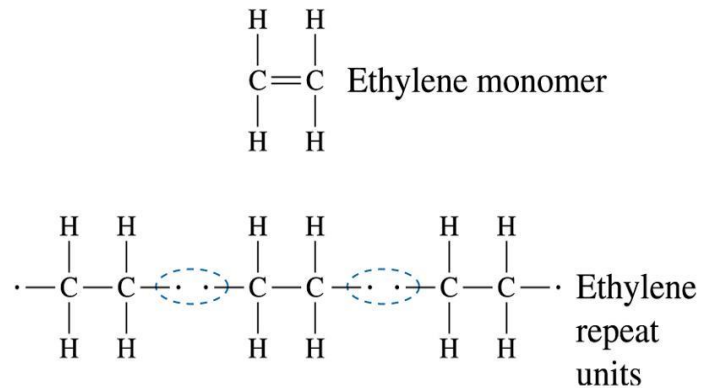
Step-reaction (or condensation)



POLYMERIZATION MECHANISM

Synthesis of polymers

Chain - reaction (or addition)



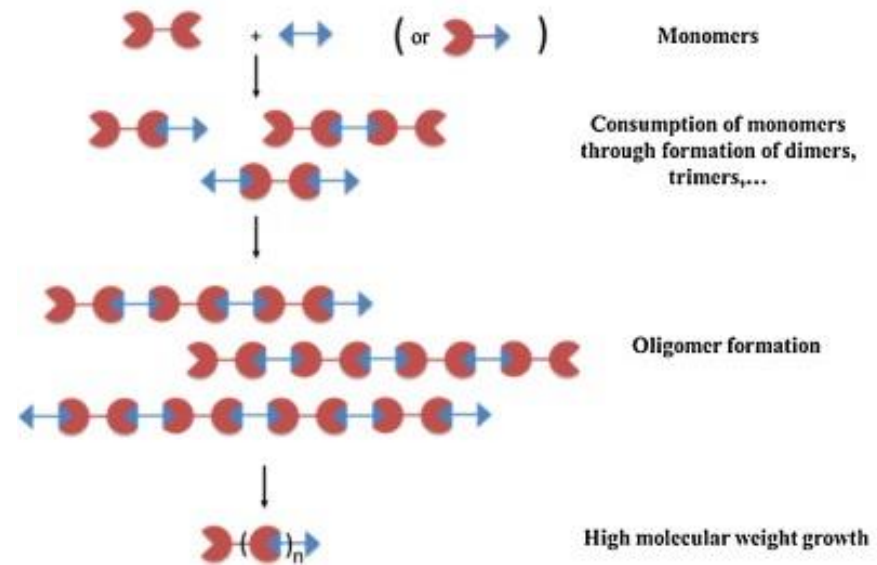
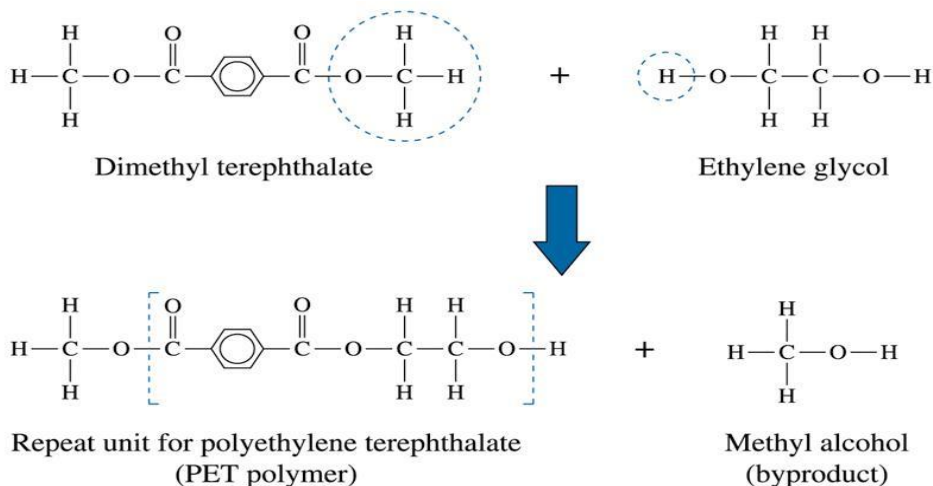
Stages:

- **Initiation:** production of free radical from initiators (benzoyl peroxide) and attachment of a polyethylene repeat unit to one of the initiator radicals
- **Propagation:** attachment of additional repeat units to propagate the chain
- **Termination:** re arrangement of the a hydrogen atom in the active ends of the chains and creation of a double covalent bond

POLYMERIZATION MECHANISM

Synthesis of polymers

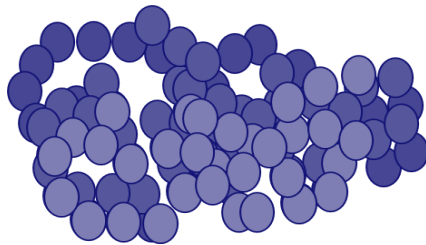
Step-reaction (condensation)



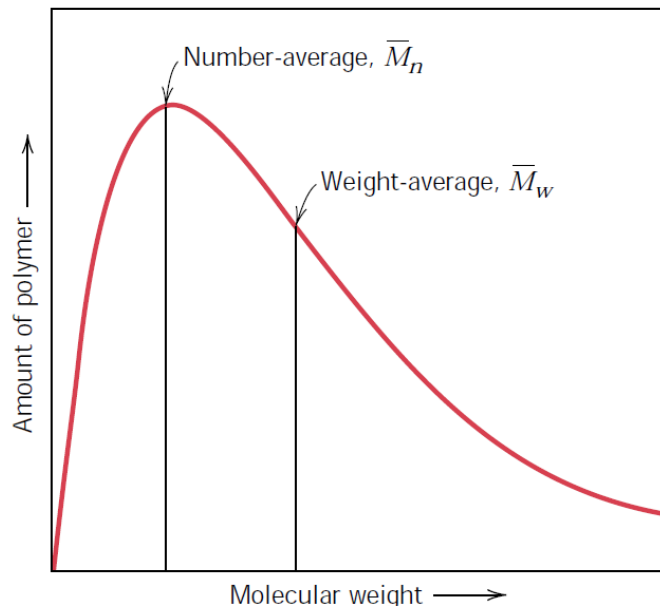
The condensation reaction for polyethylene terephthalate (PET), a common polyester. The OCH_3 group and a hydrogen atom are removed from the monomers, permitting the two monomers to join and producing methyl alcohol as a byproduct.

MOLECULAR WEIGHT

Due to the polymerization process all polymer molecules do not have exactly same molecular weight. There is a range or distribution of molecular weights. A **polymer's molecular weight** is the sum of the atomic weights of individual atoms that comprise a molecule



Degree of polymerization (X_n or DP): average number of monomer units in the polymer chain.



$$X_n = \frac{M_n}{M_o}$$

$$M_n = M_o X_n$$

where:

M_n = number-average molecular weight

M_o = monomer molecular weight

X_n = degree of polymerization in number (average number of monomer units in the polymer chain)

MOLECULAR WEIGHT

For a polymer $\Rightarrow$ distribution of chain lengths or molecular weights: $\Rightarrow$ Definition of the various molecular weights:

✓ Number average molecular weight

$$M_n = M_0 x_n = \frac{\sum W_i M_i}{\sum \frac{W_i}{M_i}} = \frac{\sum N_i M_i}{\sum N_i} = \sum x_i M_i$$

$$M_n = \sum M_i x_i \quad x_i = \frac{N_i}{\sum N_i}$$

✓ Weight average molecular weight

$$M_w = \frac{\sum W_i M_i}{\sum W_i} = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \sum w_i M_i$$

$$M_w = \sum M_i w_i$$

M_n = number-average molecular weight

M_w = weight-average molecular weight

M_i = mean molecular weight in size range i

x_i = fraction in number of molecules in range i

w_i = fraction in weight of molecules in range i

W_i = weight of molecules in range

N_i = number of molecules in range i

Index of polydispersion :

$$I = \frac{M_w}{M_n}$$

Polydispersion shows the lack of homogeneity of molecular weights

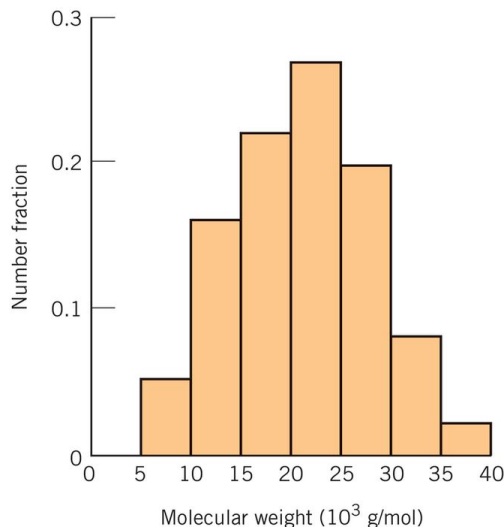
Molecular weight

Example

The number average molecular weight is obtained by dividing the chains into a series of size ranges and then determining the number fraction of chains within each size range.

a) Number-average molecular weight?

$$\overline{M}_n = \sum M_i x_i$$



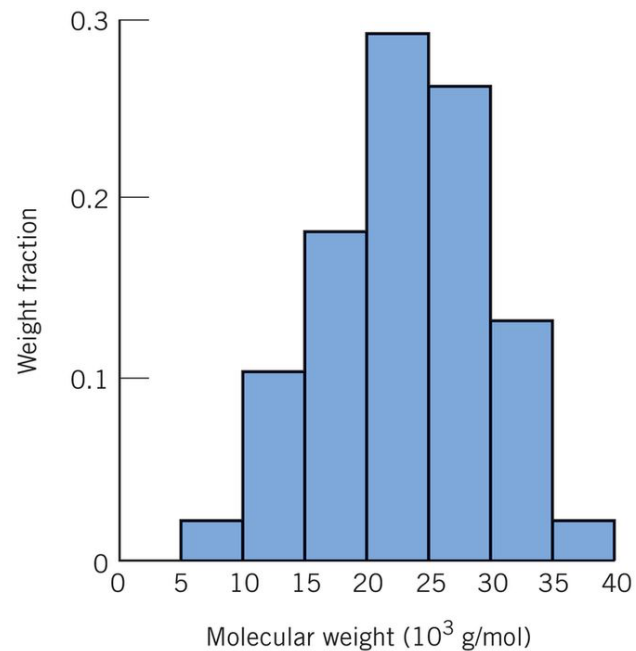
<i>Molecular Weight Range (g/mol)</i>	<i>Mean M_i (g/mol)</i>	<i>x_i</i>
5,000–10,000	7,500	0.05
10,000–15,000	12,500	0.16
15,000–20,000	17,500	0.22
20,000–25,000	22,500	0.27
25,000–30,000	27,500	0.20
30,000–35,000	32,500	0.08
35,000–40,000	37,500	0.02

Molecular weight

Example

b) Weight-average molecular weight?

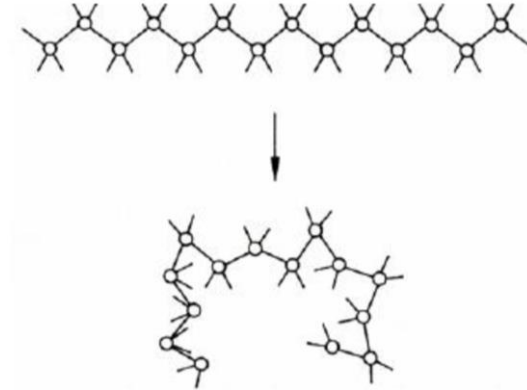
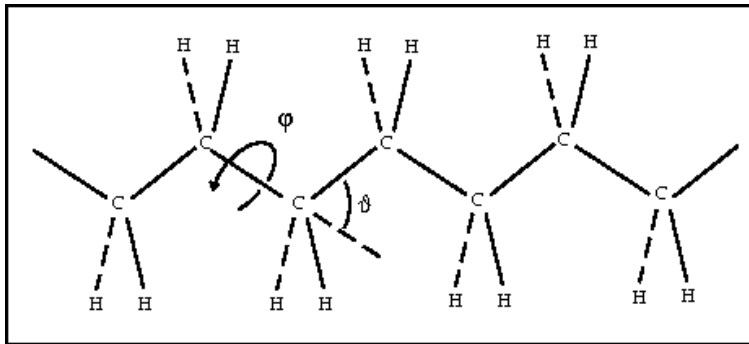
$$\overline{M}_w = \sum M_i w_i$$



<i>Molecular Weight Range (g/mol)</i>	<i>Mean M_i (g/mol)</i>	<i>w_i</i>
5,000–10,000	7,500	0.02
10,000–15,000	12,500	0.10
15,000–20,000	17,500	0.18
20,000–25,000	22,500	0.29
25,000–30,000	27,500	0.26
30,000–35,000	32,500	0.13
35,000–40,000	37,500	0.02

STRUCTURAL ASPECTS: Conformation

Polymeric molecular chains are NOT strictly straight:



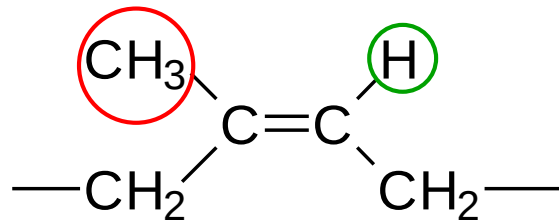
Conformation: Stereostructure of the molecule defined by its sequence of bonds and torsión angles.

Polymer chains can **change conformation by rotation** of segments along adjacent C-C axes. The **number of conformation states is limited by the type of pendant groups** in the monomers.

STRUCTURAL ASPECTS: Conformation

Conformational changes

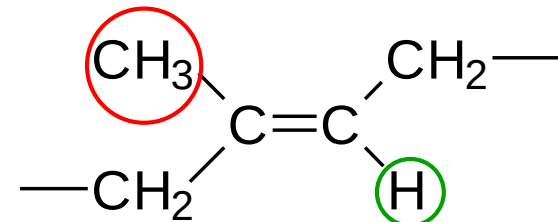
cis/trans isomerism: Chain configurations within repeat units having a double bond between chain carbon atoms:



cis

cis-isoprene
(natural rubber)

H atom and CH₃ group on
same side of chain



trans

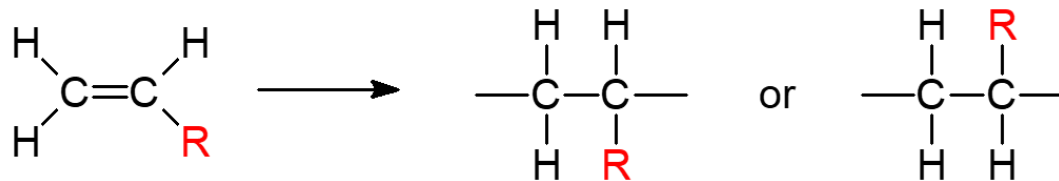
trans-isoprene
(gutta percha)

H atom and CH₃ group on
opposite sides of chain

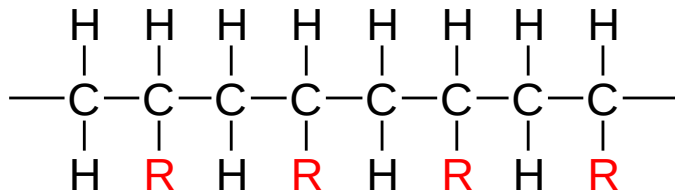
STRUCTURAL ASPECTS: TACTICITY

TACTICITY

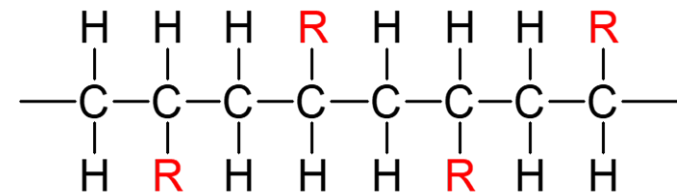
When there is spatial arrangement of **R** groups along the chain following a pattern, the polymer exhibits Tacticity (**iso** / **syndio**); if it does not follow a pattern it is **atactic**.



ISOTACTIC



SYNDIOTACTIC

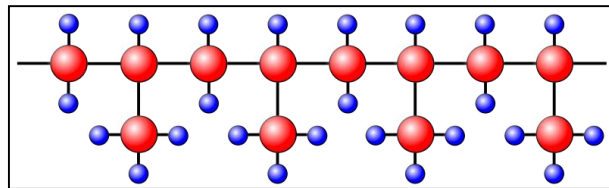


STRUCTURAL ASPECTS: TACTICITY

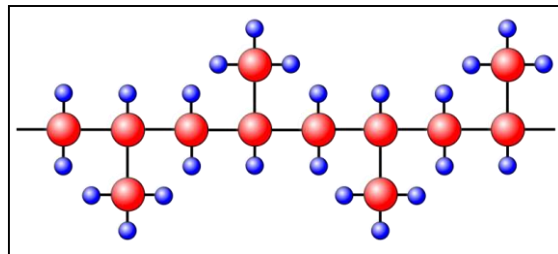
TACTICITY Examples

Polypropylene

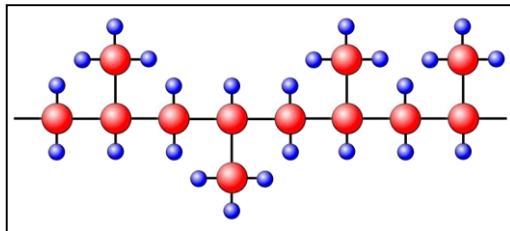
atactic polypropylene is a wax type material, with very bad mechanical properties



PP isostatic

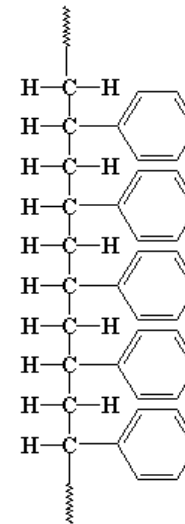


PP syndiotactic

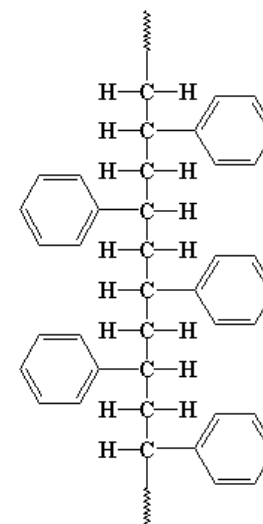


PP atactic

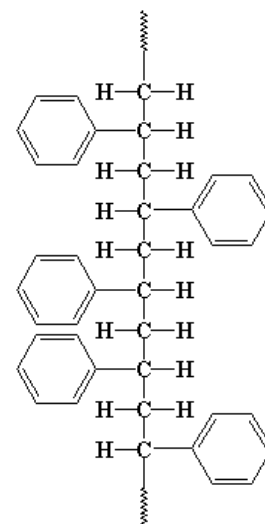
Polystyrene



Isotactic



Syndiotactic

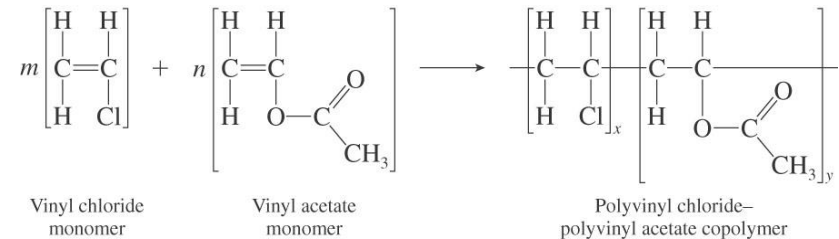


Atactic

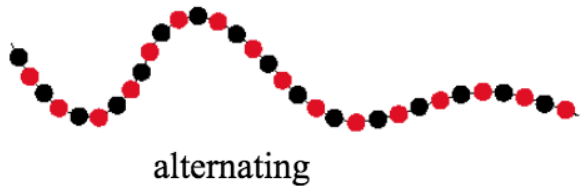
STRUCTURAL ASPECTS: Sequences

Homopolymer: a polymer containing only one type of monomer

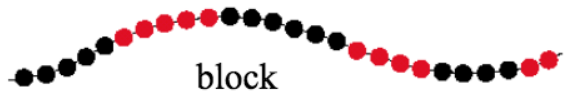
Copolymer: Two polymers polymerized together



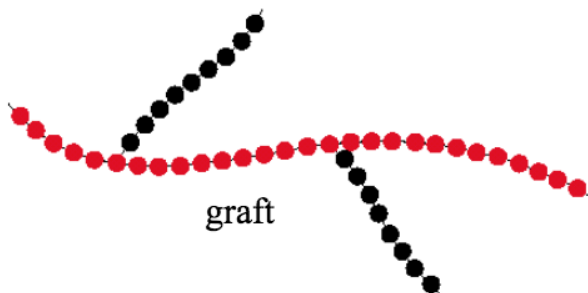
→ Copolymer with random units



→ Copolymer with alternating units

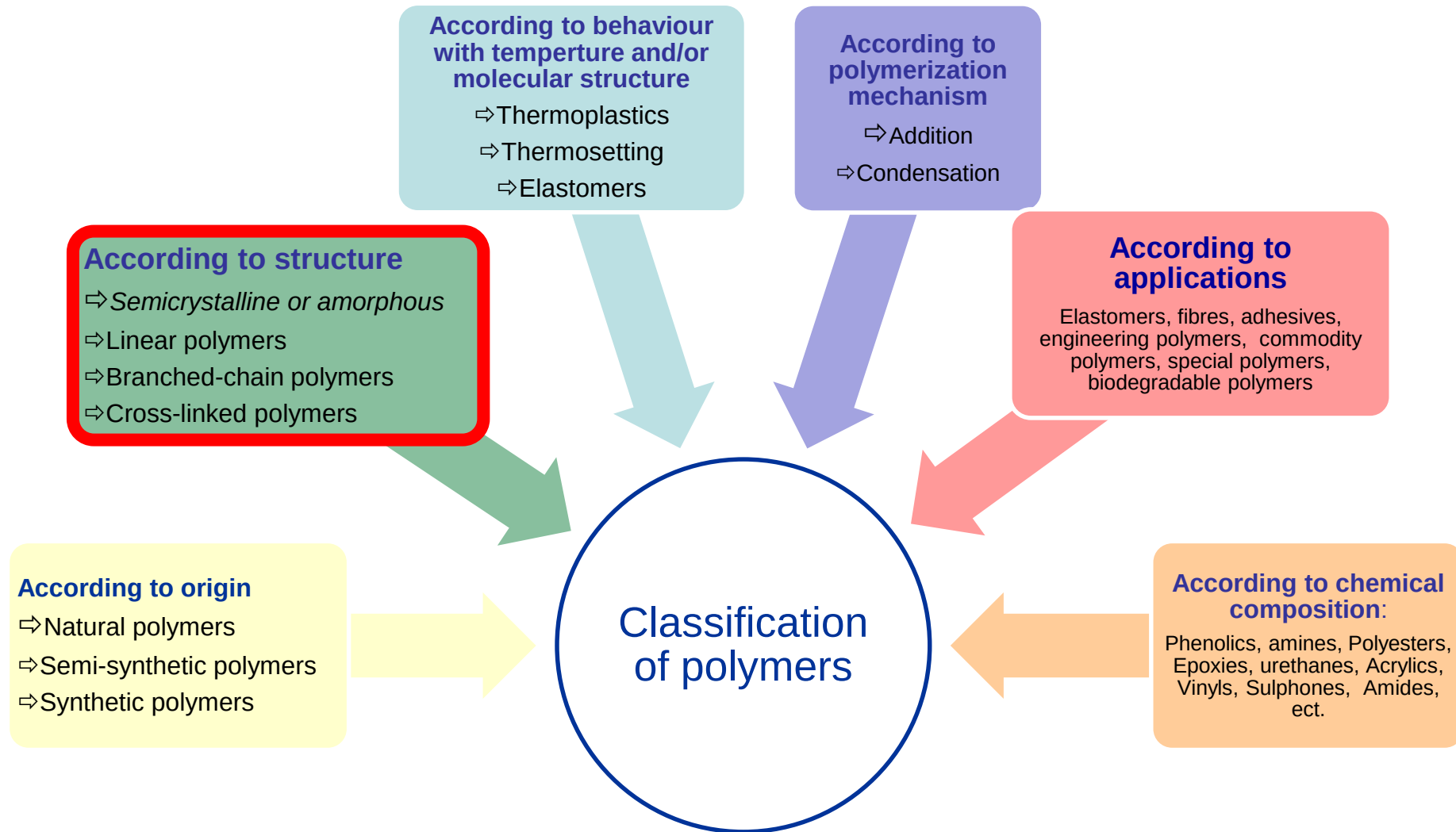


→ Copolymer with block units



→ Graft copolymers

STRUCTURAL ASPECTS

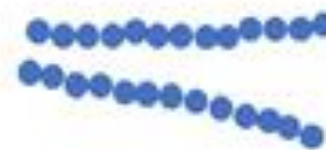


STRUCTURAL ASPECTS

MOLECULAR STRUCTURE

Linear polymers

Secondary bonds between chains



Linear Polymers



Branched Polymers

Branched polymers

Secondary bonds between chains



Crosslinked Polymers

Crosslinked polymers

Covalent bonds between chains



Networked Polymers

Network polymers

Covalent bonds between chains in 3D

Classification of polymers: Structure

The **crystallinity** of a polymer refers to the degree as to which there are regions where the **polymer chains are aligned with one another**.

a) Semicrystalline polymers

Glass transition and Fusion (glass transition temperature T_g and melting temperature T_m)

b) Amorphous polymers

Glass transition (glass transition temperature, T_g)

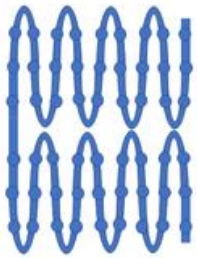
Isotropy and anisotropy

Crystallization $\uparrow \Rightarrow$ anisotropy $\uparrow$

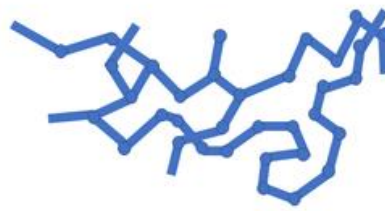
STRUCTURAL ASPECTS: CRYSTALLINITY

Semicrystalline polymers

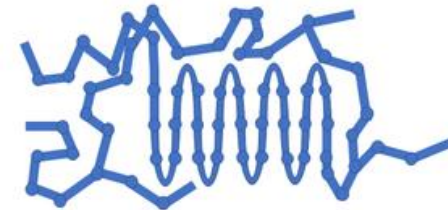
Crystalline regions (ordered) $+$ Amorphous regions (disordered) $=$ Semicrystalline polymer



Crystalline



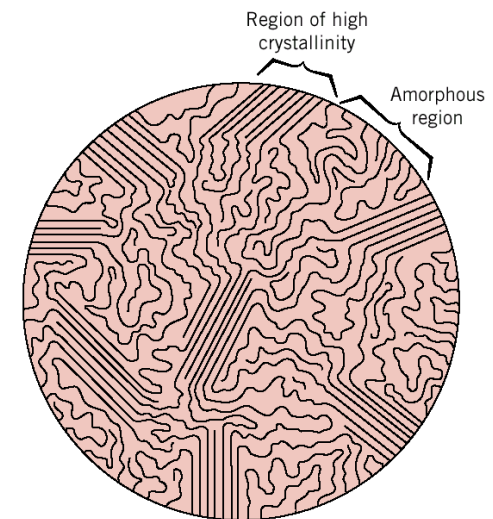
Amorphous



Semi-Crystalline

Important points:

- Properties = f (degree of crystallinity)
- $\uparrow$ Crystallinity $\Rightarrow \uparrow$ Amount of packing
- $\uparrow$ Crystallization $\Rightarrow \uparrow$ anisotropy
- ρ (crystalline polymer) $>$ ρ (amorphous polymer)

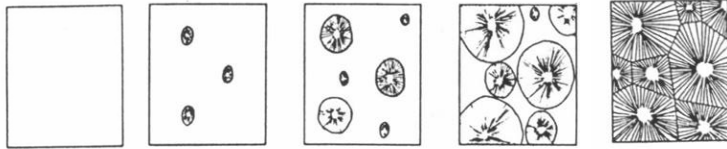


STRUCTURAL ASPECTS: CRYSTALLINITY

Semicrystalline polymers

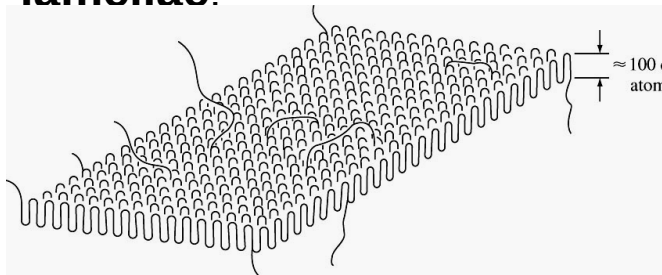
Most of polymers that crystallize from melts form **SPHERULITES**

Formed at $T < T_m$



Time, Size

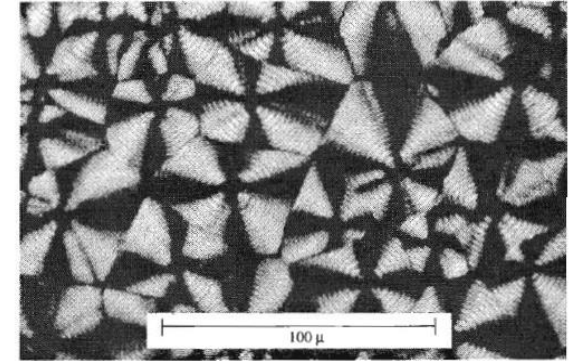
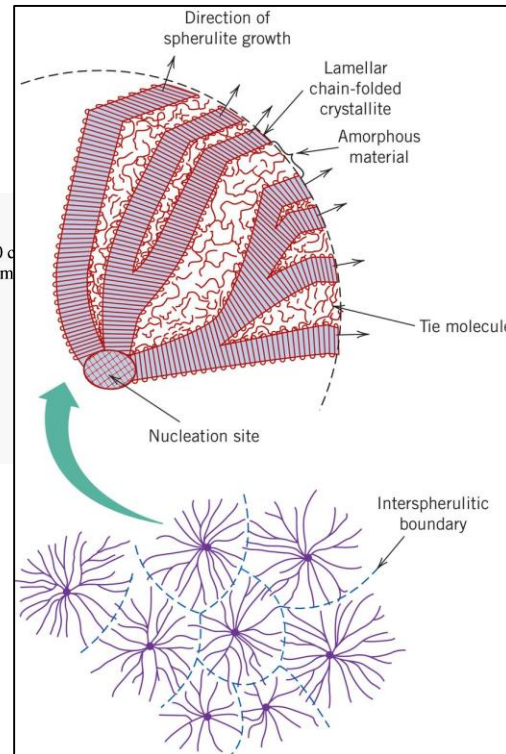
Ordered regions called **lamellae**:



Crystalline lamella



Crystalline lamella



Optical micrograph of PE with Polarized light.

$$\% \text{ crystallinity} = \frac{\rho_c (\rho_s - \rho_a)}{\rho_s (\rho_c - \rho_a)} \times 100$$

ρ_s = actual density of the polymer
 ρ_c = density of totally crystalline polymer
 ρ_a = density of totally amorphous polymer

STRUCTURAL ASPECTS: CRYSTALLINITY

Factors affecting crystallinity

✓ **Cooling rate** (if $\uparrow V_{\text{cooling}} \Rightarrow \downarrow \text{crystal size}$)

- $N^\circ \text{ crystals and size} = f(\Delta T = T_m - T_c)$

If $\Delta T \uparrow \Rightarrow \uparrow n^\circ$ and $\downarrow \text{size}$ (favours nucleation)

✓ **Structure of polymeric chains**

- Complex monomeric structures $\Rightarrow \downarrow \text{crystallinity degree}$
- Linear polymers $\Rightarrow \uparrow \text{amount of packing} \Rightarrow \uparrow \text{crystallinity degree}$
- Branched polymers $\Rightarrow \downarrow \text{crystallinity degree}$
- Polymers with asymmetric C chains:
 - Atactic $\Rightarrow$ amorphous
 - Syndiotactic, isotactic $\Rightarrow$ crystalline probably
- Linear condensation polymers $\Rightarrow$ crystalline, because they have polar groups that can form hydrogen bonds : polyamides, polyesters.

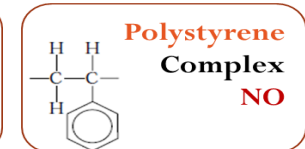
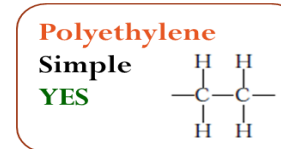
STRUCTURAL ASPECTS: CRYSTALLINITY

Factors affecting crystallinity

✓ Structure of polymeric chains

▪ Chemical structure of monomers

⇒ **Symmetric** and **simple** monomer: Greater chance for crystalline polymer



⇒ **Polarity** Polar groups usually increase crystallinity ⇒ ↑ ordering

▪ Molecular structure ⇒

structure	crystal
linear	most probable
branched	sometimes
cross-linked	almost never

▪ Copolymer sequence ⇒

Random and graft = amorphous
Alternates and block = some crystallinity

▪ Configuration states ⇒

CIS more crystalline than **TRANS**

↑ **Crystallinity** ⇒ **T_m** ↑

THERMAL TRANSITIONS

Effect of temperature

T_m : Melting temperature → crystalline regions

T_g : Glass transition temperature → amorphous regions

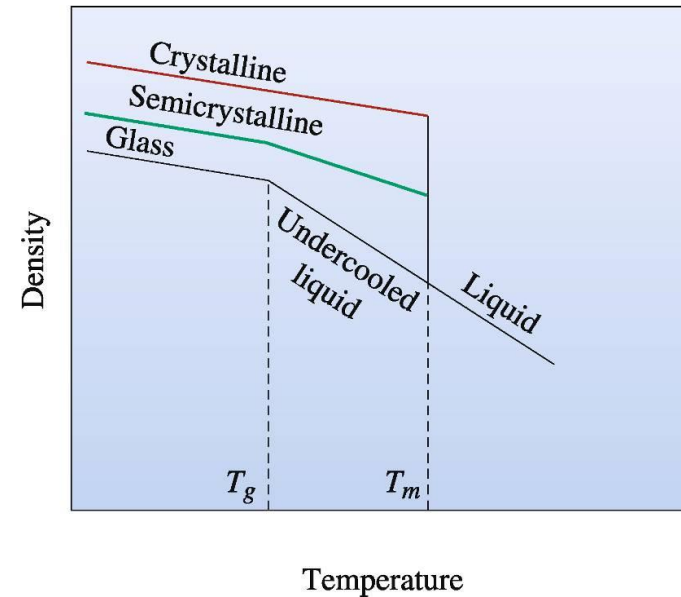
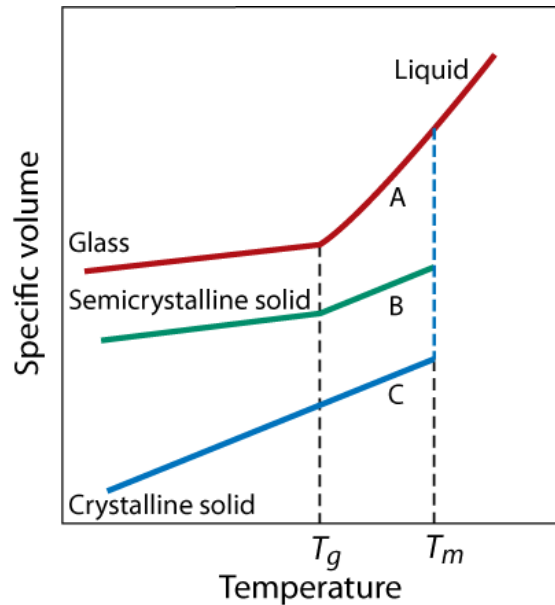


Figure 9.10

THERMAL TRANSITIONS

Factors affecting the T_g ✓ **Molecular weight**

$\uparrow M \Rightarrow \downarrow \text{Chain Mobility} \Rightarrow \uparrow T_g$

✓ **Intermolecular interactions**

$\uparrow \text{Interactions} \Rightarrow \uparrow \text{conformational energy} \Rightarrow \uparrow T_g$

✓ **Chain flexibility**

$\uparrow \text{Mobility} \Rightarrow \downarrow \text{conformational energy and } \uparrow \Delta S \Rightarrow \downarrow T_g$

✓ **Symmetry**

$\uparrow \text{Symmetry} \Rightarrow \downarrow \text{dipolar moment} \Rightarrow \downarrow T_g$

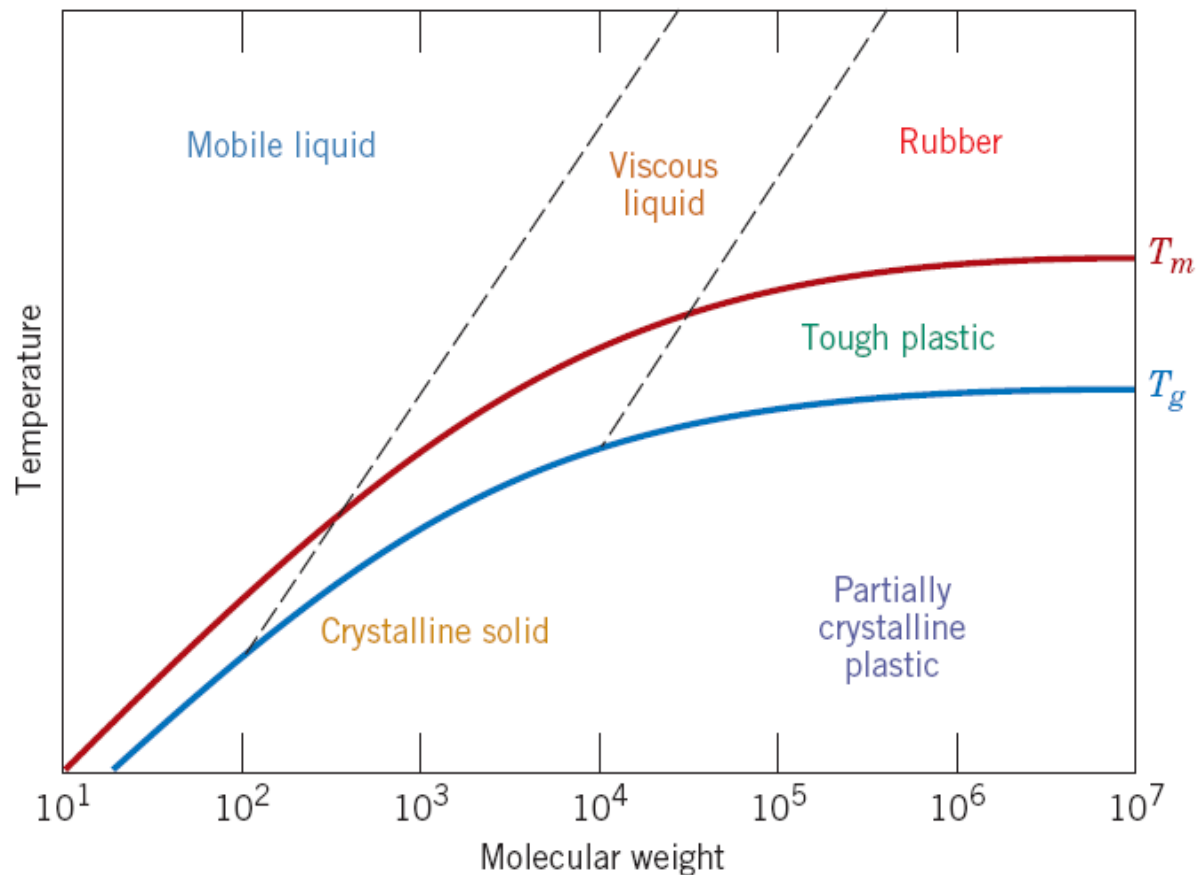
$[-\text{CH}_2-\text{CHX}-]_n$ with X	T_g (°C)
-H (HDPE)	-110
-CH ₃ (PP)	-20
-Cl (PVC)	87
-C≡N (PAN)	150
-C ₆ H ₅ (PS)	100

$[-\text{CH}_2-\text{CH}_2-]$	PE, polyethylene, $T_g = -120$ °C
$[-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-]$	PP, polypropylene, $T_g = -15$ °C
$[-\text{CH}_2-\underset{\text{C}_6\text{H}_5}{\text{CH}}-]$	PS, polystyrene, $T_g = 100$ °C
$[-\text{CH}_2-\underset{\text{N-C}_9\text{H}_7}{\text{CH}}-]$	PVK, polyvinylcarbazole, $T_g = 210$ °C

THERMAL TRANSITIONS

Factors affecting the T_g

Influence of Molecular Weight



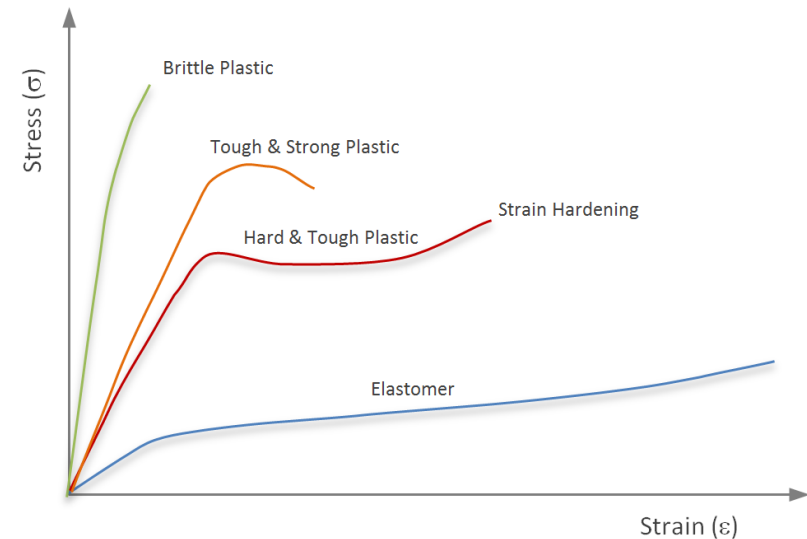
Polymeric Materials I

- Introduction.
- Classification
- Polymerization Mechanisms
- Molecular weight
- Structural aspects
- Crystallinity
- Thermal transitions: T_m and T_g .
- **Mechanical properties**
 - Stress-Strain behaviour of polymers
 - Amorphous (glassy) polymers
 - Semicrystalline polymers
 - Thermosets: Cross-linked polymers used below their T_g
 - Elastomers
 - Strength in polymers
 - Viscoelasticity
 - Relaxation modulus

MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain behaviour of polymers

- A) Brittle
- B) Tough and Ductile
- C) Tough and Ductile (strain hardening)
- D) Elastomeric

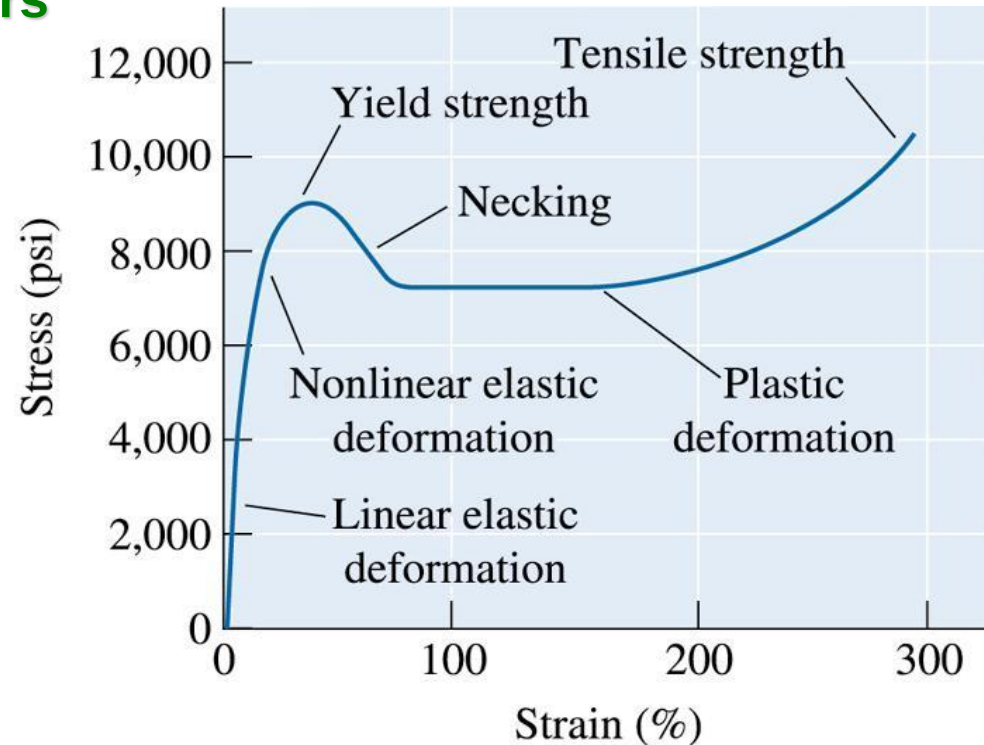
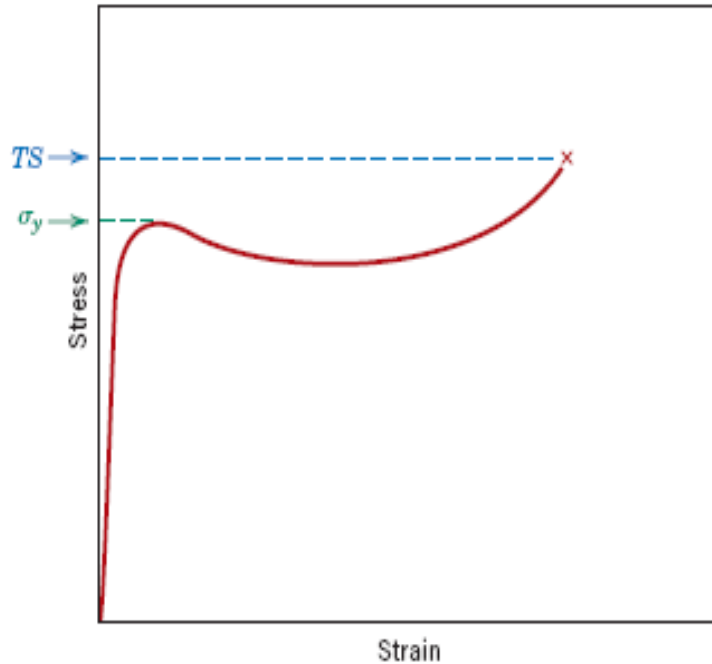


Property	Metals	Polymers
Tensile strength, σ_m	< 4100 MPa	10- 100 MPa
Elastic modulus, E	48 GPa – 410 GPa	0.007 GPa – 4 GPa
Maximum deformation, $\%\epsilon_m$	< 100%	It can reach more than 1000%

$$\text{Mechanical properties} = f(T, v_{\text{def}}, \text{environment})$$

MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers



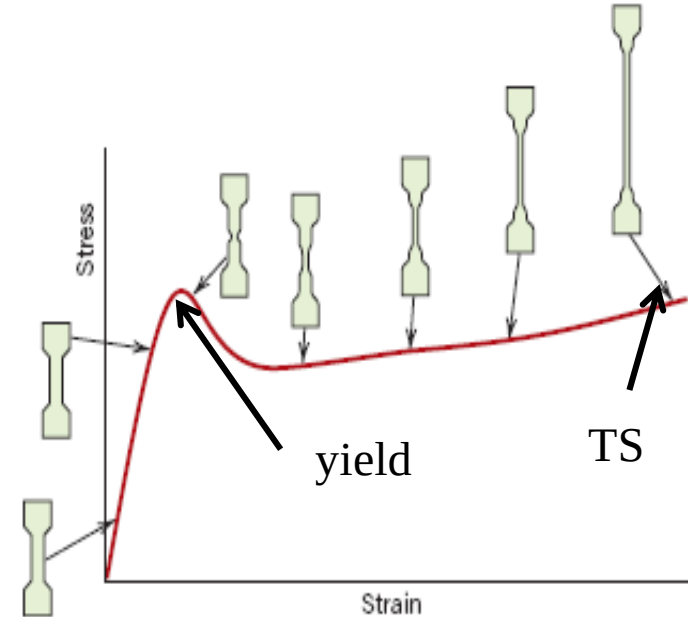
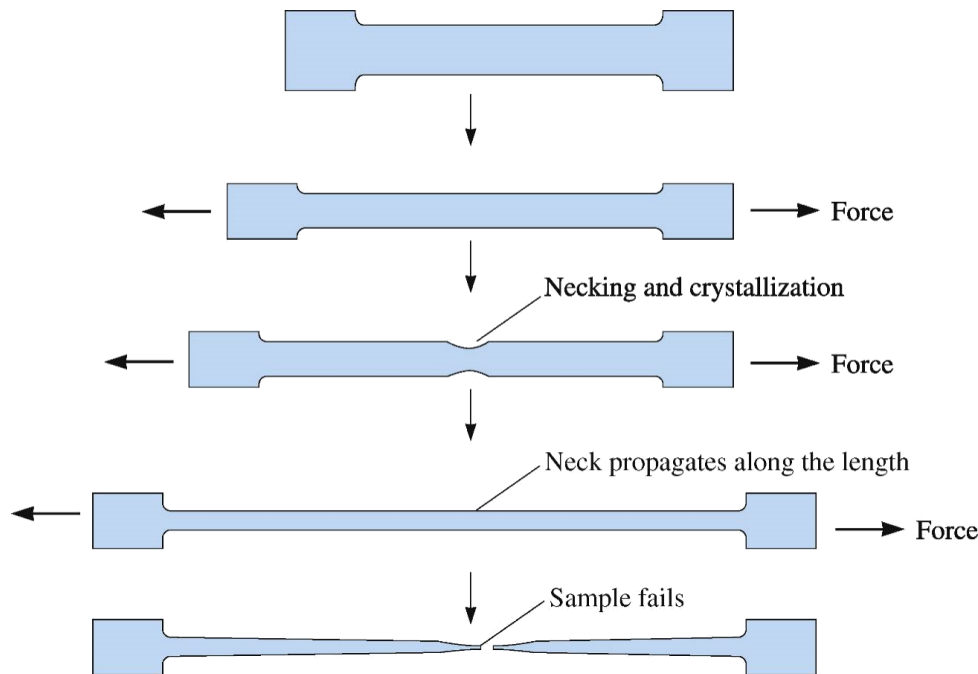
- Young's Modulus (or Modulus of elasticity) determined in the same way as for metals
- Yield strength (s_y): stress at the termination of the elastic region (maximum).
- Tensile strength: stress at which fracture occurs

The stress-strain curve for 6,6-nylon, a typical thermoplastic polymer.

MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers

Mechanisms of Plastic Deformation



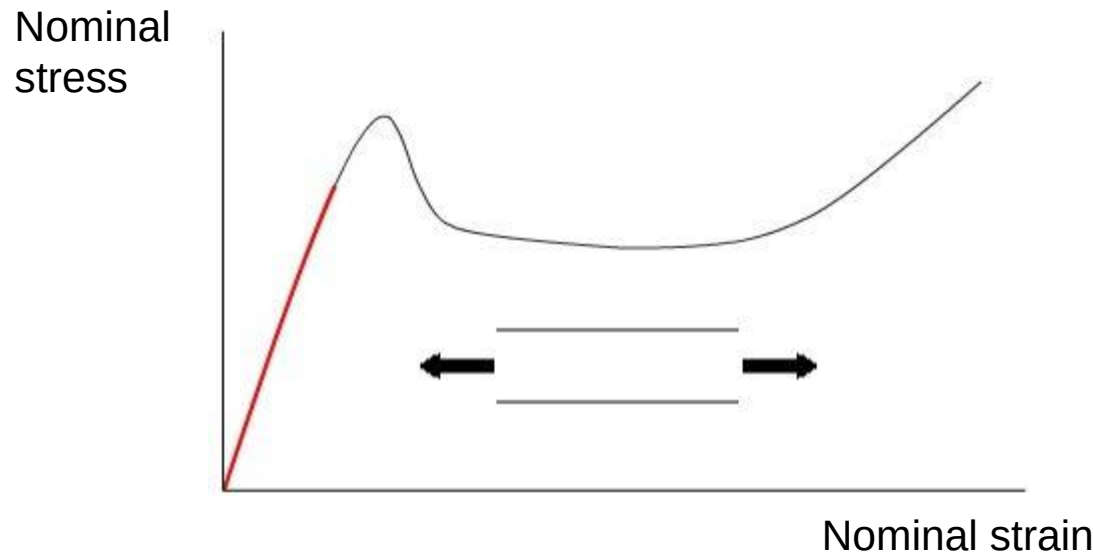
- Segments of the chain are aligned, necking appears
- Increases the strength in the zone where the chains are aligned
- Extends necking instead of making it sharper
- Necking is extended and the chains ordered

MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers

Mechanisms of Plastic Deformation

Elastic Region. In this region stress is proportional to strain and the deformation is completely **reversible**; Hooke's law is obeyed. The gradient of the straight-line section of the curve gives the elastic modulus of the material. In practice the elastic region of the curve is **unlikely to be perfectly straight**; the elastic modulus is taken as the **tangent to the first part of the curve**.

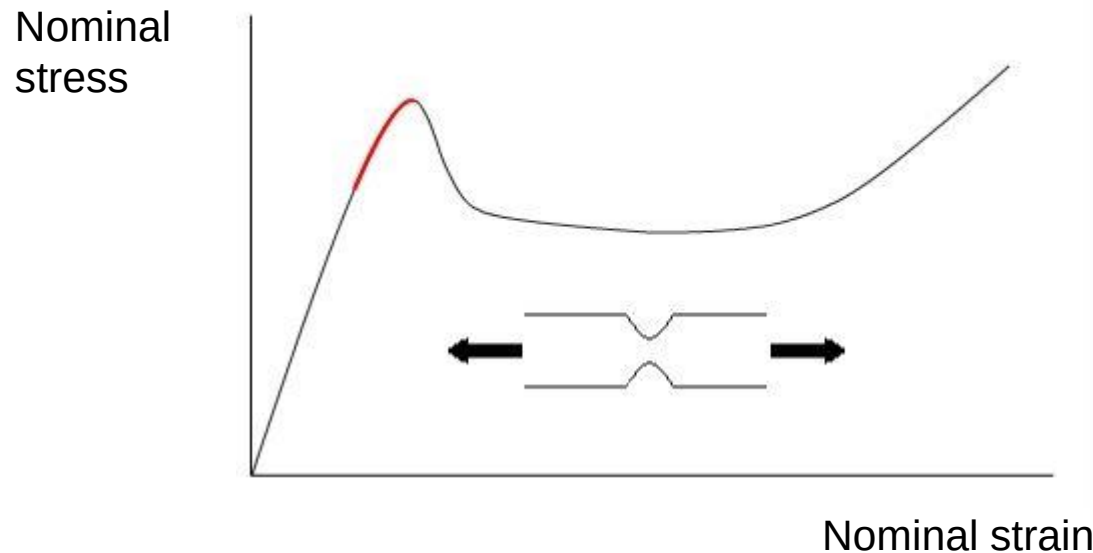


MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers

Mechanisms of Plastic Deformation

Yield. The yield stress of the polymer is poorly defined as the stress-strain curve bends over at the top of the elastic region. The value at the end of the elastic region or at the top of the curve could be used; either would be adequate **but the value at the top of the curve is generally used.** For a typical ductile polymer yield may occur at 5 - 10 % strain, whereas, in contrast, a metal generally yields at less than 0.1 % strain.

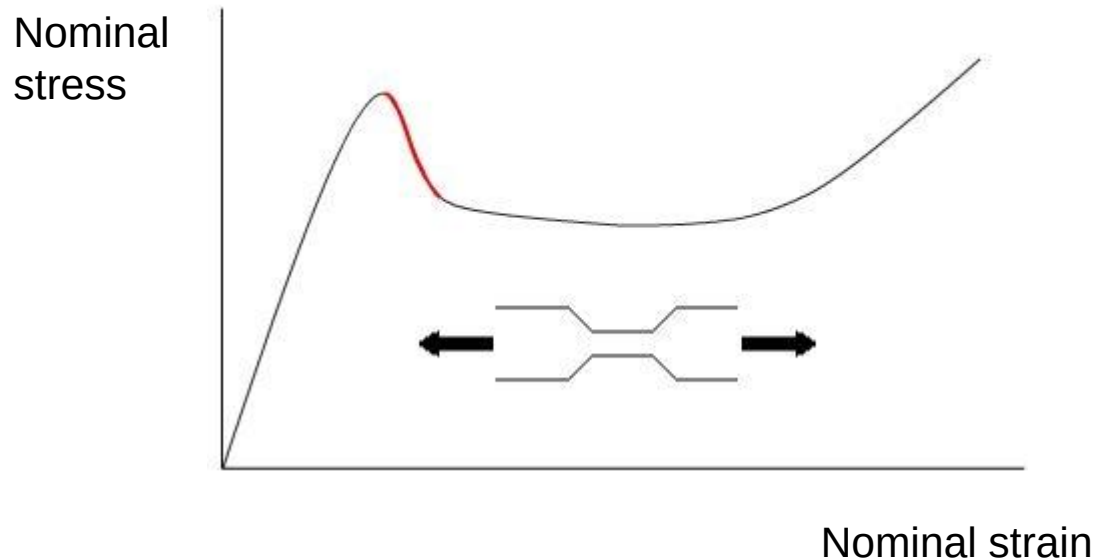


MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers

Mechanisms of Plastic Deformation

Necking. At the yield point there is a local decrease in cross-sectional area at a point along the length of the sample. **Necking involves a balance between work hardening and softening.** As the stress-strain curve shows, the **nominal stress falls after yield**, before settling at a constant value as the neck extends along the sample.

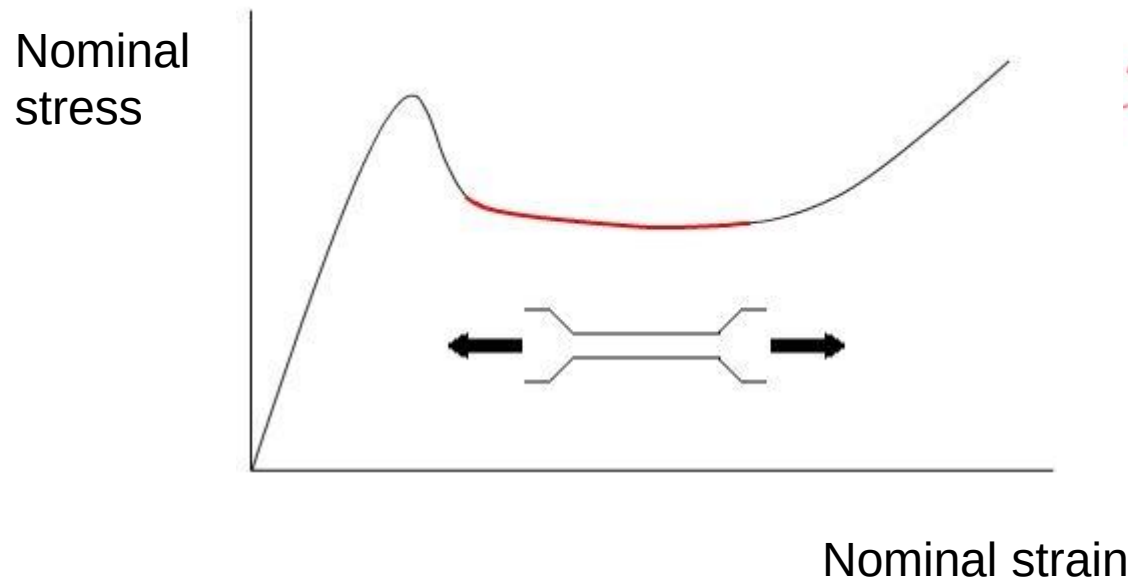


MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers

Mechanisms of Plastic Deformation

Cold drawing. The process by which the **neck extends** is known as strain hardening or cold drawing. The polymer chains unravel, aligning themselves parallel to the direction of the applied stress.

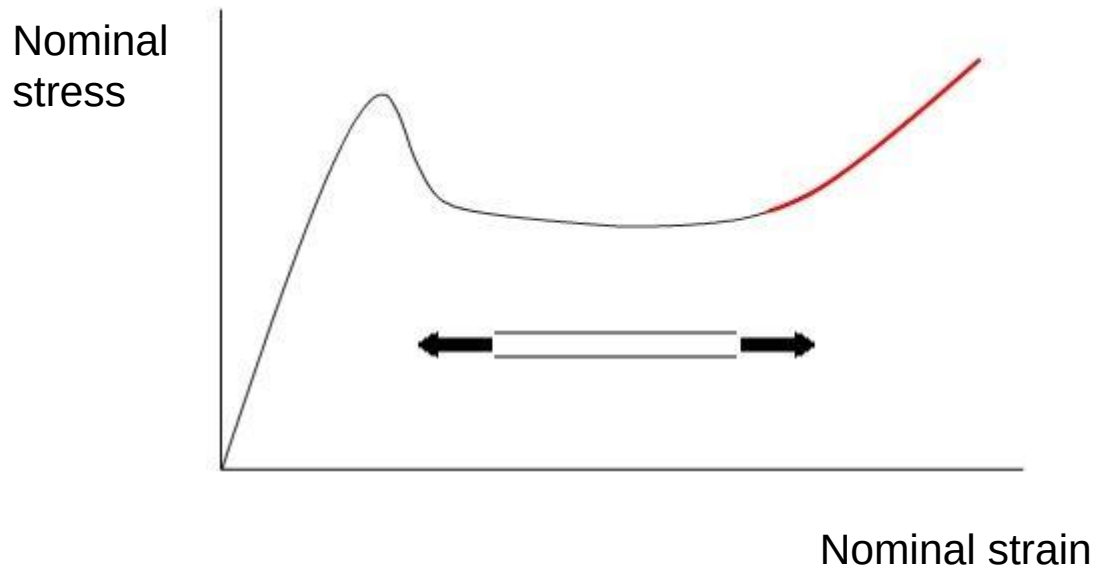


MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers

Mechanisms of Plastic Deformation

Strain Hardening. The increase in the final section of the curve is a result of strain hardening, which occurs once the whole sample is necked. The stress rises until fracture takes place. **Strain hardening is principally a consequence of chain orientation.** Molecules are aligned parallel to the stretching direction in the cold drawn regions of both amorphous and crystalline polymers. alignment strengthens the necked region and reduces its rate of deformation. Oriented polymer is significantly stronger and stiffer than an isotropic one.

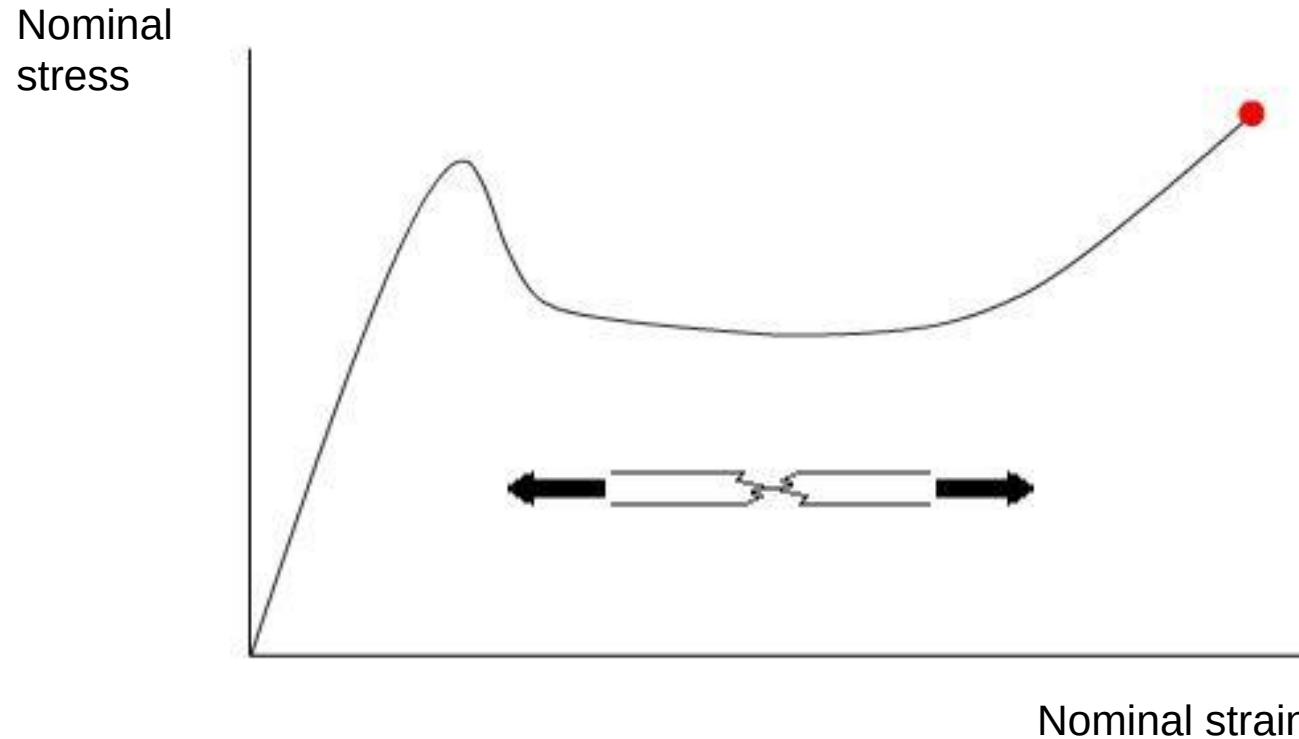


MECHANICAL PROPERTIES OF POLYMERS

Stress-Strain curves in polymers

Mechanisms of Plastic Deformation

Failure. When no further chain alignment is possible, the sample will fracture.



MECHANICAL PROPERTIES OF POLYMERS

Mechanisms of Plastic Deformation



Elastic deformation



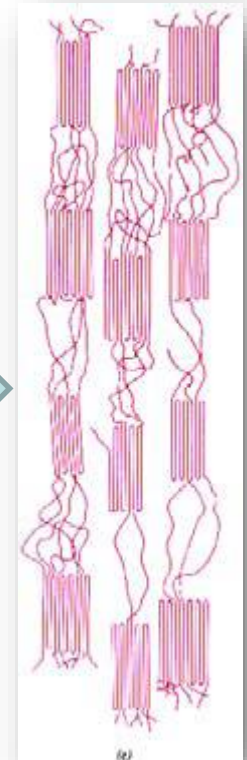
Elongation of amorphous tie chains



Tilting of lamellar chain folds



Separation of crystalline block segments and alignment with tensile axis



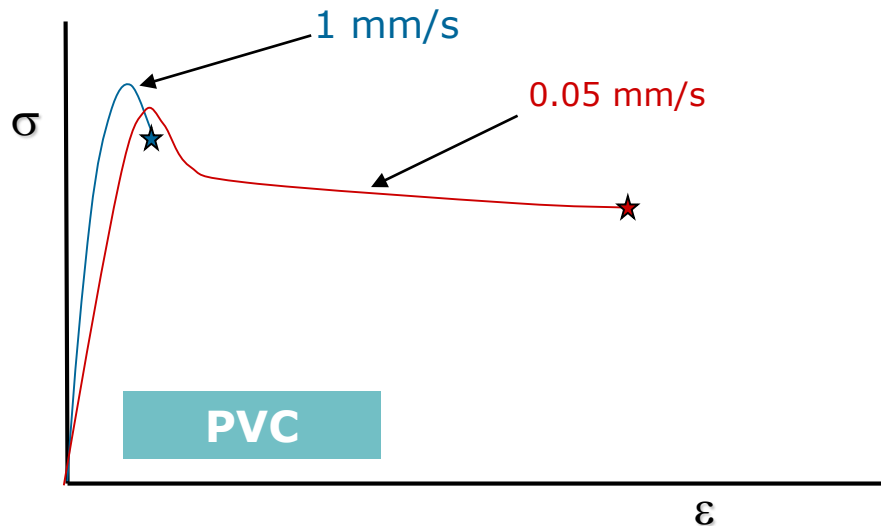
Orientation of both block segments and tie chains with tensile axis

MECHANICAL PROPERTIES OF POLYMERS

Major factors affecting strength are **temperature** and **strain rate**:

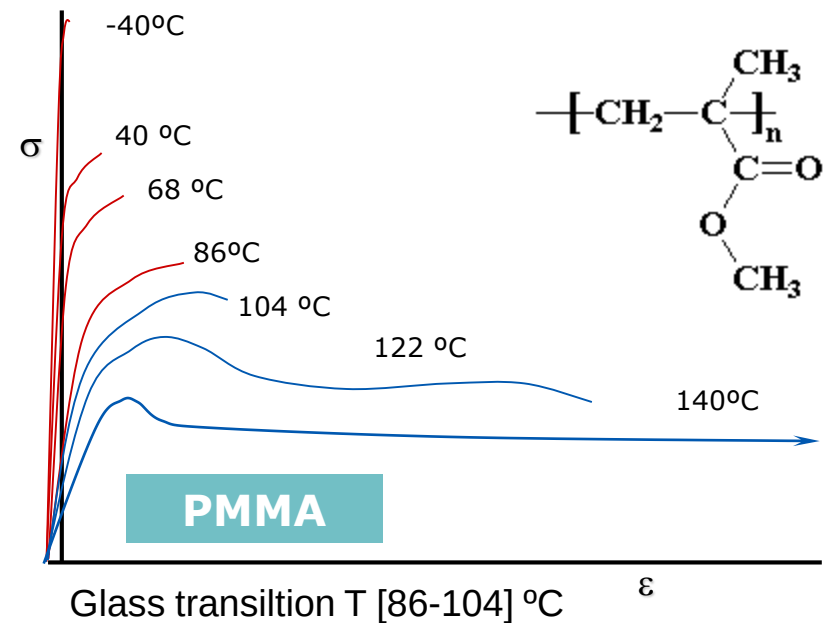
Strain rate

Strain rate $\uparrow$ {
 $\uparrow$ Brittleness
 $\downarrow$ Percent deformation at fracture
 $\uparrow$ Difficulty of disentanglement



Temperature

If $T \uparrow$ {
 $\downarrow$ Modulus of elasticity
 $\downarrow$ Tensile strength
 $\uparrow$ Percent deformation to fracture



If the strain rate is high $\Rightarrow$ brittle behavior (difficult to unravel chains)

MECHANICAL PROPERTIES OF POLYMERS

Other factors that influence mechanical behaviour

- ✓ The tensile strength of the polymer increases with increase in **molecular weight** ($\Rightarrow$ *high number of entanglements*).

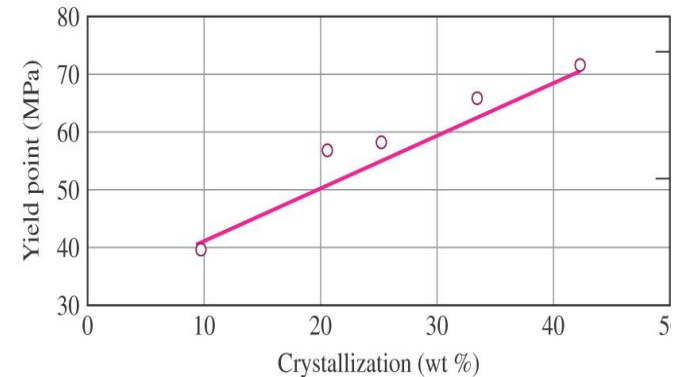
Long chains $\Rightarrow$ *high number of entanglements*

$\uparrow M : \uparrow \sigma, E$

- ✓ The strength of the polymer also increases with its **degree of crystallinity**.

$\rho(\text{crystalline polymer}) > \rho(\text{amorphous polymer})$

$\uparrow \% \text{ crystallinity} \Rightarrow \boxed{\uparrow \sigma, E}$



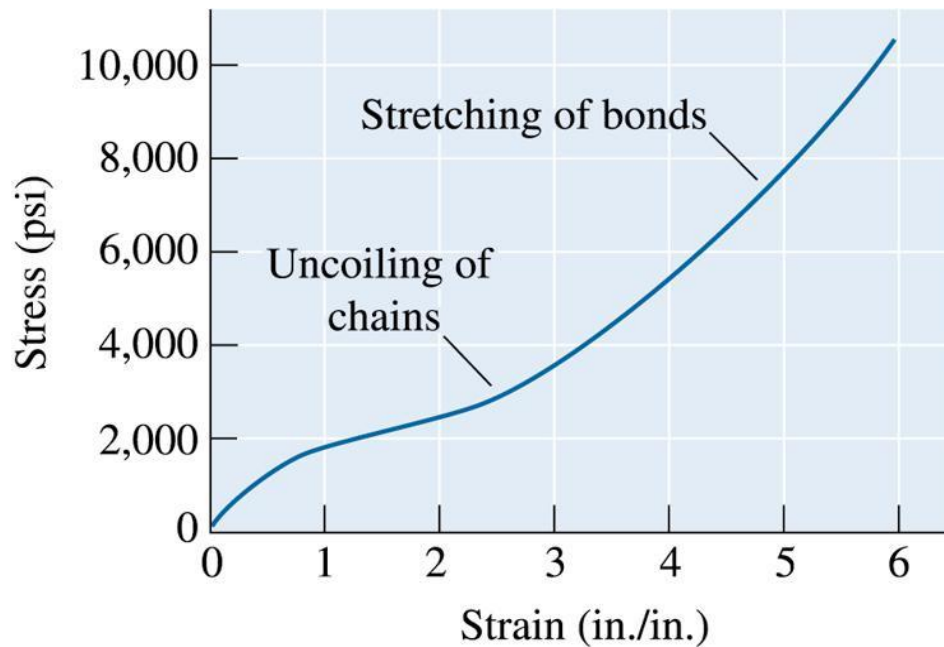
- ✓ Strength can be increased by **increasing the degree of crosslinking** (thermosetting polymers) $\Rightarrow$ *inhibits chain motion - makes it more brittle*.
- ✓ Deforming a polymer can increase its strength: *chains become oriented leading to more effective interactions between them*.
- ✓ Introduction of lateral groups:
 - Voluminous groups $\Rightarrow \uparrow$ stiffness, σ , $\downarrow$ ductility
 - Introduction of polar groups $\Rightarrow \uparrow$ intermolecular forces $\Rightarrow \uparrow \sigma$

PVC	40-80 MPa
PE	17-36 MPa

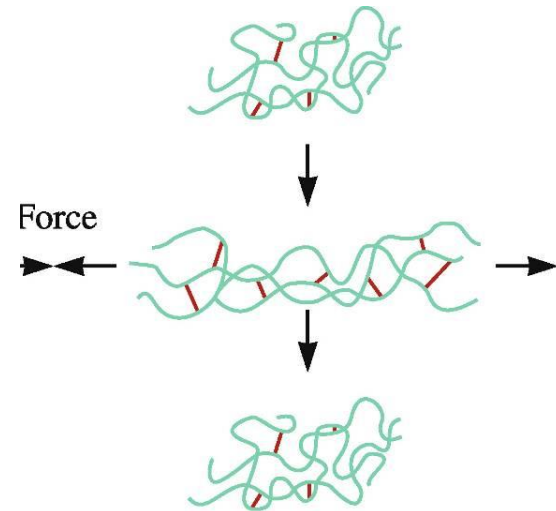
MECHANICAL PROPERTIES OF POLYMERS

Elastomers

The stress-strain curve for an **elastomer**: Virtually all of the deformation is **elastic**; therefore, the modulus of elasticity varies as the strain changes



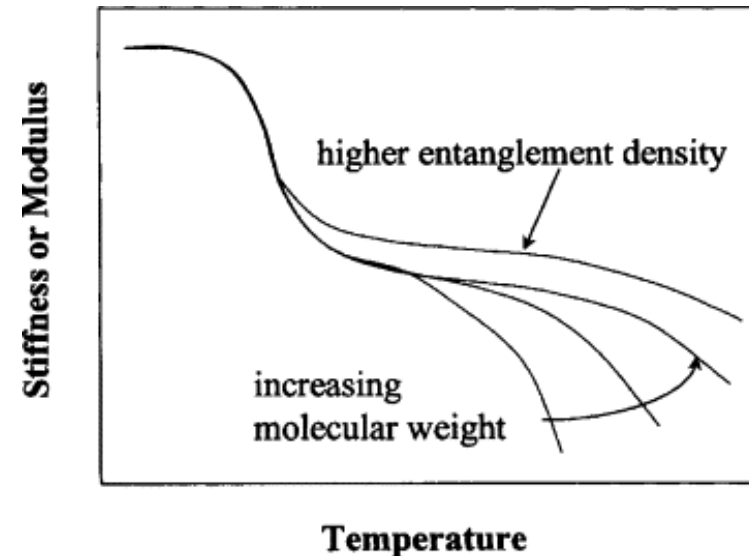
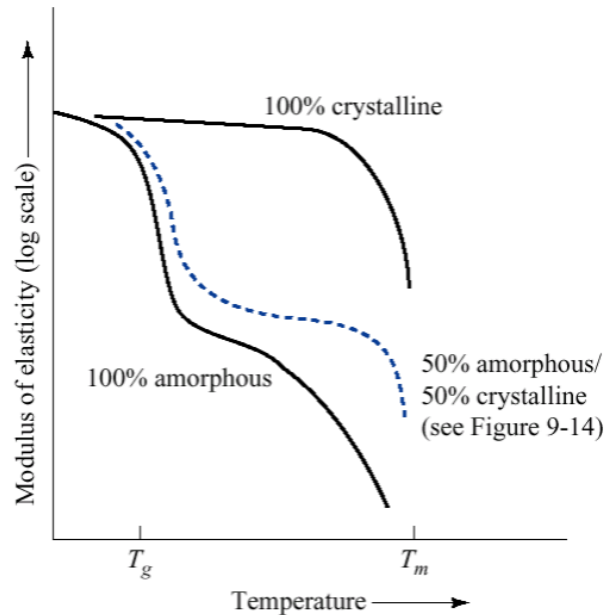
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MECHANICAL PROPERTIES OF POLYMERS

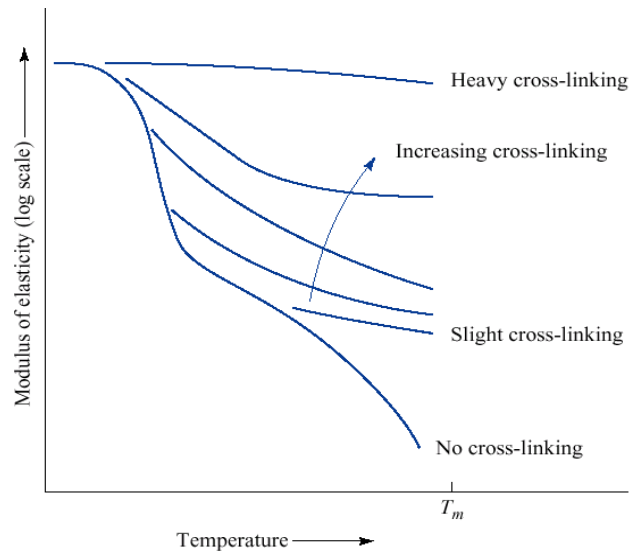
Strength in polymers

Effect of the crystallinity (thermoplastic polymers) and MW in the modulus of elasticity:

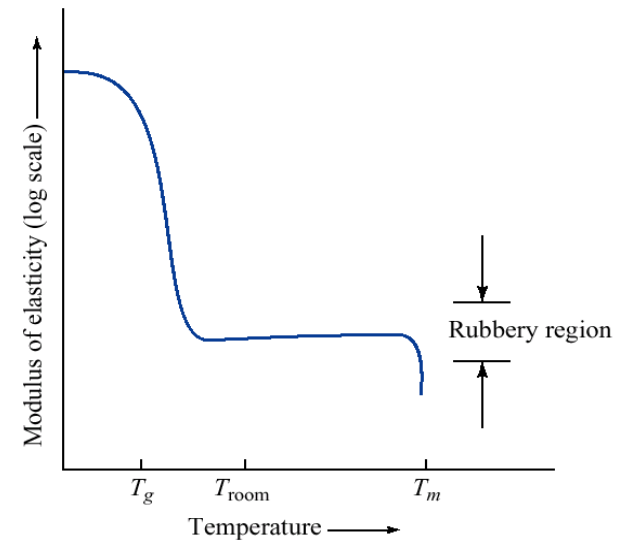


MECHANICAL PROPERTIES OF POLYMERS

Increasing crosslinking (**Thermosetting polymers**) leads to an increase of the stiffness



Effect of temperature on an **Elastomer**: pronounced rubbery region



MECHANICAL PROPERTIES OF POLYMERS

Viscoelastic functions. Creep and relaxation

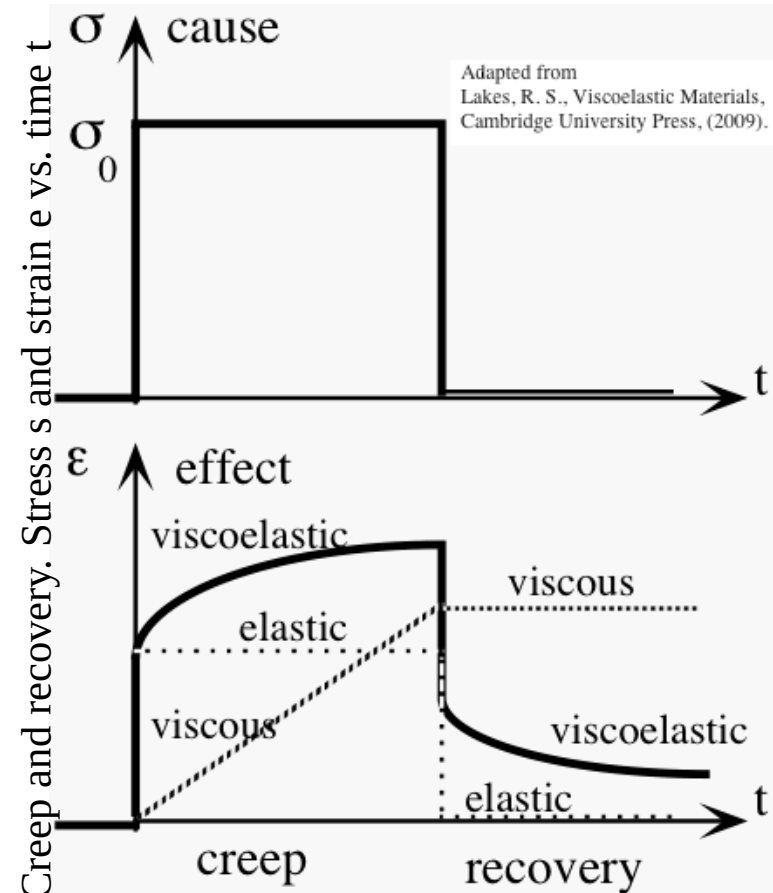
Creep is a slow, progressive deformation of a material under constant stress. In one dimension, suppose the history of stress σ as it depends on time t to be a step function beginning at time zero:

$$\sigma(t) = \sigma_0 H(t)$$

$H(t)$ is the unit Heaviside step function defined as zero for t less than zero, one for t greater than zero and for $t = 0$.

The strain $\varepsilon(t)$ in a viscoelastic material will increase with time. The ratio

$J(t) = (\varepsilon(t) / \sigma_0)$ is the **creep compliance**



MECHANICAL PROPERTIES OF POLYMERS

Stress relaxation is the gradual decrease of stress when the material is held at constant strain. If we suppose the strain history to be a step function beginning at time zero:

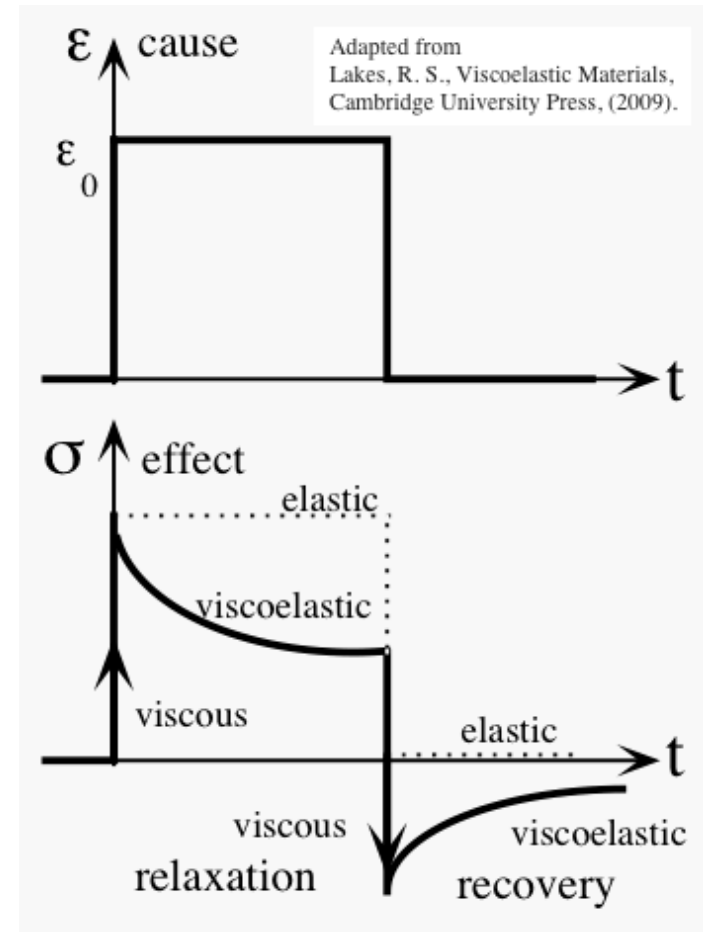
$$\varepsilon(t) = \varepsilon_0 H(t)$$

The ratio

$E(t) = (\sigma(t) / \varepsilon_0)$ is called the **relaxation modulus**

The relaxation modulus depends on temperature.

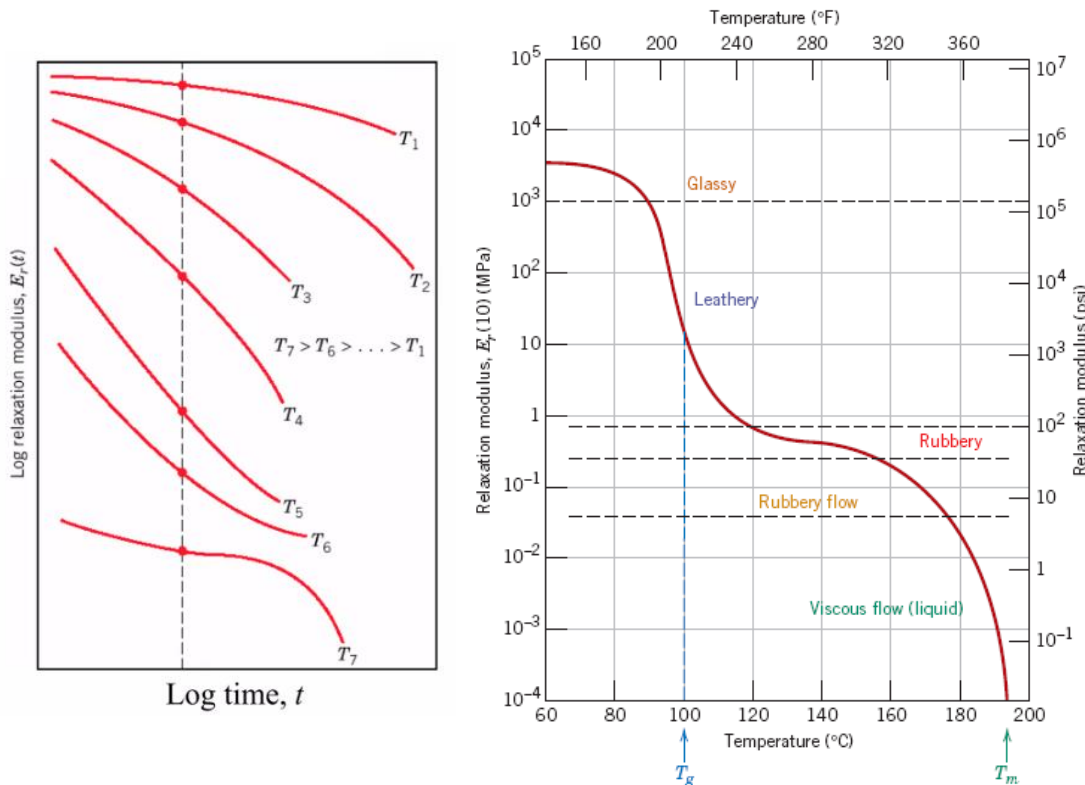
The rate with which the stress is applied also influences the viscoelastic characteristics. *An increase of this rate exerts the same effect of a decrease of temperature.*



a) Applied strain and b) induced stress as functions of time for a viscoelastic material.

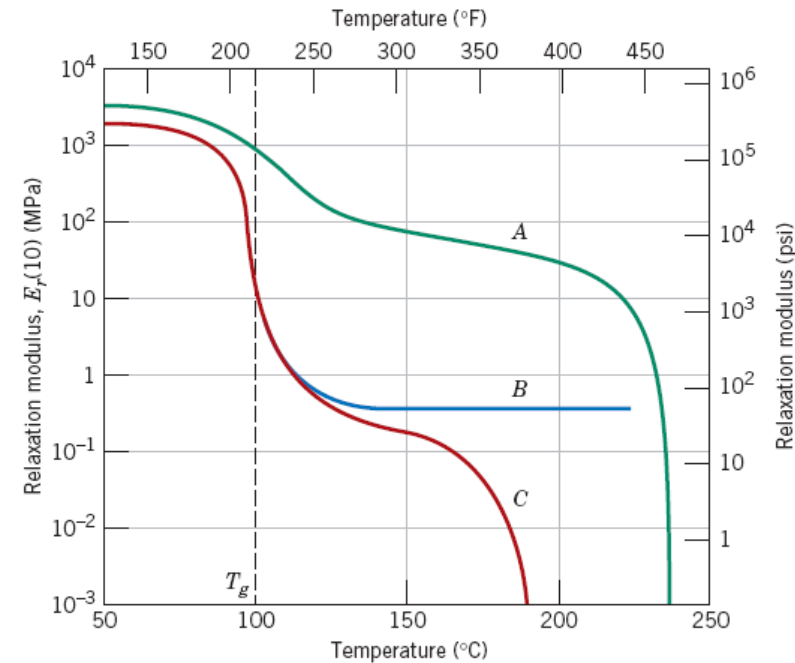
MECHANICAL PROPERTIES OF POLYMERS

Relaxation modulus



Logarithm of the relaxation modulus (at a certain time of being applied the stress, $t_1 = 10\text{s}$) vs. Temperature. Atactic polystyrene.

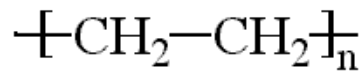
polystyrene.



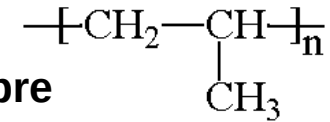
- A) crystalline isotactic
- B) lightly crosslinked atactic
- C) amorphous

EXAMPLES

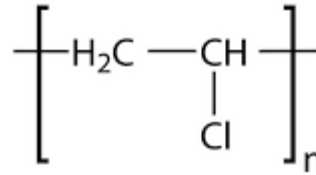
Poly ethylene PE:
Most popular plastic



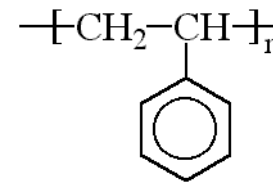
Polypropylene PP:
used as plastic and as fibre



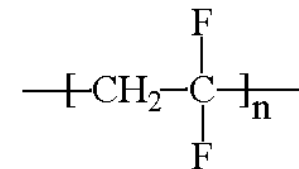
Poly (vinyl chloride) PVC:
pipes, construction, cables



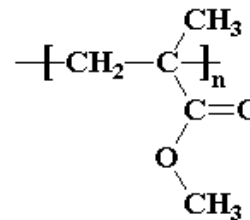
Polystyrene PS:
economic and resistant. Styrofoam™: foam of PS



Poly (vinylidene fluoride) PVDF: high electric and fire resistance



Polymethyl methacrylate PMMA:
transparent plastic. Substitute of crystal



Nylon (polyamides):
most common
polymers used as fibers

